

Algebra 1
Unit 10 Assessment

Name _____
Date _____
Period _____

1. The function $G(x) = |0.05x - 0.85| + 1.77$ models the gas price, in dollars, for January 6, 2020 to August 24, 2020, where x is the number of weeks since January 6, 2020.

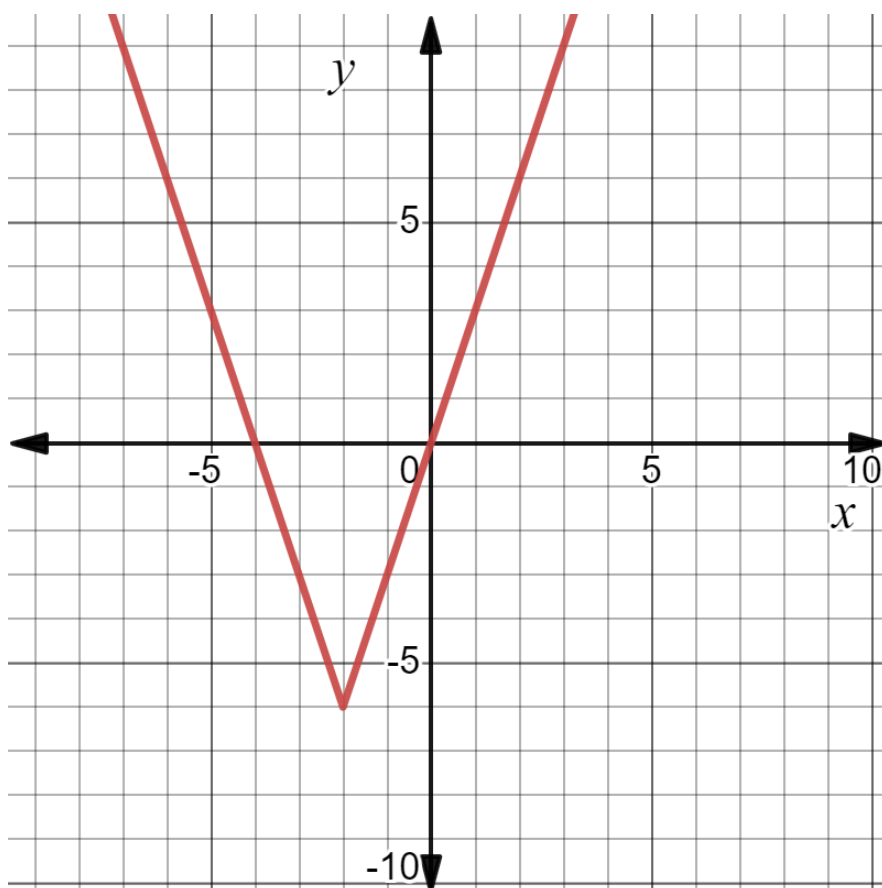
Determine when gas prices were \$2.52.

work	
Number of weeks since January 6, 2020	2 and 32

2. An equation is shown.

$$y = 3|x + 2| - 6$$

Use the coordinate grid to graph the equation.



Determine the solution for each equation. Show your work.

3. $|6 - 4k| - 30 = 0$

$k =$

-6 and 9

4. $|8x - 11| + 45 = 6$

$x =$

No solutions

5. $|n + 6| = 19$

$n =$

13 and -25

6. $-3|y + 5| + 27 = 0$

$y =$

4 and -14

7. The circumference of a regulation discus used in the men's Olympic track and field events should be 691.15 millimeters (mm). The manufacturing tolerance is within 3.15 mm. Write an absolute value equation that could be used to determine the maximum and minimum value of x , the circumference of a regulation discus used in the men's Olympic track and field events.

Equation	$ x - 691.15 = 3.15$
-----------------	-----------------------

8. A table of values for the function $f(x) = -\frac{1}{2}|x - 4|$ is shown.

x	-6	-3	-2	0	3	5	6	9
y	-5	-3.5	-3	-2	-0.5	-0.5	-1	-2.5

- a. Complete the table.

Name	Key Feature	Name	Key Feature
x -intercepts	$(4, 0)$	increasing	$x < 4$
y -intercept	$(0, -2)$	decreasing	$x > 4$
vertex	$(4, 0)$	positive	none
axis of symmetry	$x = 4$	negative	$x < 4$ or $x > 4$
domain	$\mathbb{R}$	range	$-\infty < y \leq 0$

- b. Use the coordinate grid to graph the equation.

